Factors Contributing to Emergency Care among Cardiac Patients and Strategies for Improving Rapid Response in the Kandy District

Leo Club of University of Peradeniya

Abstract

Cardiac emergencies demand immediate and well-coordinated medical intervention, as delays in treatment can significantly increase the risk of complications and mortality. In the Kandy district, patients frequently experience critical delays in accessing timely emergency care due to a combination of individual, social, and systemic factors. These delays often occur within the crucial "golden hour," where early intervention is most effective in improving survival outcomes. This study aims to investigate the key factors contributing to such delays and to identify strategies for improving rapid response in cardiac emergencies within this regional context. A mixed-methods research design was employed to provide a comprehensive understanding of the issue. Quantitative data were collected through a structured questionnaire administered to 30 participants, focusing on variables such as symptom awareness, response behavior, perceived delays, and attitudes toward technological interventions. The data were analyzed using IBM SPSS software to generate descriptive statistics and correlation analyses, particularly examining relationships between awareness levels, emotional responses, and delays in seeking treatment. In addition, qualitative data were gathered through 10 semi-structured interviews, allowing for an in-depth exploration of patient experiences, communication barriers, and vulnerabilities during emergency situations. A thematic analysis approach was used to identify recurring patterns and key themes. The findings reveal that lack of symptom awareness, fear and confusion, and physical inability to communicate during an emergency are major contributors to delayed response. A strong positive correlation was identified between emotional uncertainty and delays in seeking medical care, indicating that psychological factors play a critical role in patient decision-making. Furthermore, individuals living alone were found to be at significantly higher risk due to their dependence on external assistance during emergencies. Logistical challenges, including transportation delays and limited immediate access to emergency medical services, further compound these issues. Importantly, the study also highlights a high level of acceptance toward technological solutions among participants. The majority expressed strong willingness to adopt wearable health monitoring devices and emphasized the need for faster communication systems linking patients, families, and healthcare providers. Based on these findings, the study suggests that integrating continuous monitoring technologies with automated alert mechanisms can serve as an effective strategy to reduce response times and improve emergency care outcomes. In conclusion, this research underscores the urgent need to transition from reactive emergency response models to more proactive and technology-supported systems. By addressing both human and systemic limitations, such strategies have the potential to significantly enhance the efficiency of cardiac emergency care and ultimately save lives in the Kandy district

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Keywords: Cardiac emergencies, emergency response delay, wearable technology, smartwatch, continuous health monitoring, IoT healthcare, symptom awareness, communication barriers, vulnerable populations, Kandy District, Sri Lanka.

Introduction

Cardiovascular diseases continue to be one of the leading causes of death worldwide, accounting for a significant proportion of global mortality. Among these, acute cardiac events such as heart attacks require immediate medical attention to prevent severe complications or death. The concept of the "golden hour" emphasizes the importance of receiving timely treatment within the first hour of symptom onset, as early medical intervention can drastically improve survival rates. However, despite advancements in healthcare systems, many patients still experience critical delays in accessing emergency care, particularly in developing regions.

In the context of Sri Lanka, and specifically within the Kandy district, several factors contribute to delays in emergency cardiac care. Geographic and infrastructural challenges, such as rural terrain and traffic congestion, can significantly slow down the transportation of patients to healthcare facilities. In addition, limitations in emergency medical services and delays in ambulance availability further extend the time taken to receive treatment. These systemic issues are often compounded by individual-level factors, including lack of awareness of cardiac symptoms, fear, confusion, and misinterpretation of early warning signs.

One of the most critical challenges identified in emergency response systems is their reliance on a reactive model, where the patient or a bystander must recognize the severity of the situation and initiate a call for help. This approach is inherently flawed in cases where patients are physically incapacitated, uncertain about their symptoms, or living alone without immediate support. In such situations, valuable time is lost before any medical intervention can begin, reducing the effectiveness of treatment and increasing the risk of mortality.

Furthermore, social factors such as age, education level, and living conditions play a significant role in influencing how individuals respond to cardiac emergencies. Elderly individuals and those living alone are particularly vulnerable, as they may lack the physical ability or external assistance required to seek help promptly. Similarly, individuals with lower levels of education may have limited awareness of cardiac symptoms, leading to delayed decision-making and increased risk.

Given these challenges, there is a growing need to explore strategies that can enhance the speed and efficiency of emergency response systems. While traditional approaches focus on improving ambulance services and hospital readiness, there is increasing attention on the role of technology in bridging the gap between symptom onset and medical intervention. Technological advancements, particularly in wearable health monitoring and real-time communication systems, offer promising opportunities to address these delays by enabling continuous monitoring and faster alert mechanisms.

This study aims to examine the factors affecting emergency care response among cardiac patients in the Kandy district and to identify practical strategies for improving rapid response. By combining quantitative data analysis with qualitative insights, the research seeks to provide a comprehensive understanding of both human and systemic barriers to timely care. Ultimately, the findings of this study will contribute to the development of more effective and innovative approaches to managing cardiac emergencies, with the potential to improve patient outcomes and reduce mortality rates.

Literature Review

Rapid and effective medical intervention remains the most critical factor in improving survival outcomes during cardiac emergencies. Despite advancements in healthcare systems, delays in receiving timely treatment continue to pose a significant global challenge, particularly in developing regions. This literature review examines the key factors contributing to delays in emergency cardiac care, including patient-related, systemic, and technological aspects, while also exploring potential strategies to improve rapid response.

1.The "Golden Hour" and Treatment Delays

The concept of the "golden hour" is widely recognized in cardiovascular medicine as the critical period immediately following the onset of a cardiac event, during which prompt medical intervention significantly increases the likelihood of survival. According to Morrison et al. (2019), early treatments such as percutaneous coronary intervention (PCI) can reduce mortality rates by up to 30% when administered within this time frame. However, several studies indicate that pre-hospital delays remain a major barrier to achieving optimal outcomes.

Research by Anand et al. (2022) highlights that in low- and middle-income countries, delays in emergency response are often prolonged due to limited pre-hospital care infrastructure and insufficient emergency medical services. While these studies emphasize the importance of timely intervention, they also reveal that systemic improvements alone are insufficient without addressing patient-level delays.

2.Patient Awareness and Decision-Making Behavior

A significant proportion of emergency response delay is attributed to patient-related factors, particularly the lack of awareness and misinterpretation of cardiac symptoms. Studies consistently show that many individuals fail to recognize early warning signs such as chest discomfort, shortness of breath, or fatigue, often attributing them to less serious conditions. As a result, patients adopt a "wait and see" approach, which leads to critical time loss.

Furthermore, research indicates that patient decision time can account for up to 75% of total pre-hospital delay. Emotional factors such as fear, denial, and confusion further exacerbate this issue, preventing individuals from seeking immediate medical attention. This suggests that

improving awareness alone may not be sufficient; psychological and behavioral factors must also be addressed to reduce delays effectively.

3.Communication Barriers and Social Vulnerability

Another major factor influencing emergency response is the presence of communication barriers during a cardiac event. In many cases, patients become physically incapacitated, limiting their ability to call for help or communicate their condition effectively. This highlights a critical weakness in traditional emergency response systems, which rely heavily on the patient's ability to initiate action.

Social vulnerability further intensifies this problem. Elderly individuals and those living alone face significantly higher risks due to the absence of immediate assistance. Even when symptoms are recognized, the lack of a support system can delay the initiation of emergency care. These findings demonstrate that emergency response is not solely a medical issue but also a social one, requiring solutions that account for varying living conditions and support networks.

4.Geographical and Logistical Constraints

In addition to patient-related challenges, geographical and infrastructural factors play a crucial role in delaying emergency care. Studies conducted in the Sri Lankan context, such as those by Perera and Silva (2023), highlight the impact of rural terrain, traffic congestion, and limited availability of emergency transport services on response times. These logistical barriers often result in significant delays between the moment help is requested and when the patient reaches a healthcare facility.

Moreover, the lack of pre-arrival communication between patients and hospitals limits the ability of healthcare providers to prepare for incoming emergencies. This further reduces the efficiency of treatment upon arrival, emphasizing the need for systems that can provide real-time patient information before hospital admission.

5.Emerging Role of Technology in Cardiac Care

Recent advancements in healthcare technology have introduced new possibilities for addressing delays in emergency response. The integration of the Internet of Things (IoT) and wearable devices has enabled continuous health monitoring and real-time data collection. According to Spaccarotella et al. (2020), modern wearable devices equipped with electrocardiogram (ECG) capabilities can effectively detect cardiac abnormalities, including arrhythmias and signs of acute coronary syndromes.

These technologies offer a shift from traditional reactive healthcare models toward proactive monitoring systems. By continuously tracking physiological data, wearable devices have the potential to identify anomalies before symptoms become severe, thereby reducing dependence on patient recognition and manual intervention. However, despite their potential, the adoption and integration of such technologies within local healthcare systems remain limited.

Research Gap

Although existing literature extensively discusses the medical importance of rapid intervention and the factors contributing to emergency delays, there is a noticeable gap in integrating these insights into practical, technology-driven solutions within specific local contexts. Many studies focus either on clinical outcomes or technological capabilities in isolation, without addressing how these solutions can be adapted to real-world social and logistical conditions.

In particular, there is limited research exploring how continuous monitoring systems can overcome both human and systemic barriers simultaneously in regions like the Kandy district. Additionally, the level of patient acceptance and readiness to adopt such technologies in local settings remains underexplored. Therefore, this study aims to bridge this gap by examining both the contributing factors to delays and the potential for innovative strategies to improve rapid emergency response.

Significance of the Study

This research is vital as it addresses a critical knowledge and technological gap in managing cardiac emergencies within the Kandy area. While existing studies heavily document the medical consequences of treatment delays, fewer explore the integration of automated, real-time IoT solutions tailored to local logistical and social constraints. By identifying the exact barriers patients face—such as poor symptom recognition, fear, and the vulnerabilities of living alone—this study provides the necessary evidence to justify a Smartwatch and Web Interface alert system. The findings will not only contribute to the academic understanding of pre-hospital delays but also guide the practical development of a 24/7 monitoring solution designed to drastically reduce response times, improve hospital preparedness, and ultimately save lives.

Problem Statement

Cardiac patients frequently experience delays in receiving emergency medical care due to a combination of poor symptom awareness, communication barriers during the onset of an attack, and logistical challenges. When a cardiac event occurs, patients are often physically unable to call for help, and those living alone face an amplified risk of mortality. Furthermore, hospitals receive no prior data regarding the patient's real-time vitals before they arrive at the emergency room. These delays and information gaps reduce the effectiveness of medical interventions. Therefore, it is critical to understand these specific barriers to propose a technology-driven, continuous monitoring solution that automates emergency alerts.

Objective of the Study

To identify the primary factors causing delays in emergency care for cardiac patients and to evaluate the impact of symptom awareness and fear on delays in seeking treatments.

Research Questions

1. What are the primary factors contributing to delays in emergency medical responses for cardiac patients in the Kandy district?
2. How does a patient's symptom awareness and fear or confusion affect their decision time during a cardiac emergency?
3. To what extent do logistical barriers and social vulnerabilities (such as living alone or the inability to communicate) hinder rapid emergency response?
4. What is the level of patient acceptance and willingness to adopt wearable IoT technology for continuous health monitoring?
5. How can a synchronized smartwatch and web interface system mitigate existing delays, notify families, and improve hospital preparedness?

Methodology

- **Study Deign**

  This study adopts a pragmatic research philosophy, utilizing a mixed-methods approach to gain a comprehensive understanding of the challenges cardiac patients face. Pragmatism is highly appropriate for systems development research, as it combines measurable statistical data with real-world user experiences to justify a practical technological solution.

- **Sampling and Data Collection**

  The research incorporates both quantitative and qualitative data:

  Quantitative: A structured survey was administered to 30 participants. Data was collected to measure awareness, perceived delays, the need for a solution, and technology acceptance.

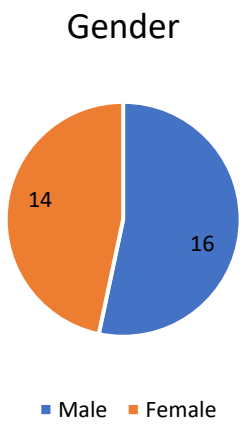
  Qualitative: Semi-structured, in-depth interviews were conducted with 10 participants to explore personal experiences regarding emergency communication, fear, and vulnerability.

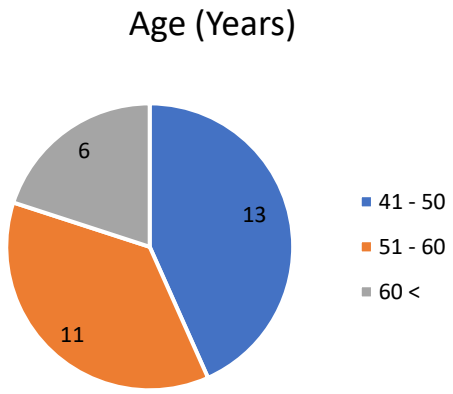
- **Data Analysis and Sampling Techniques**

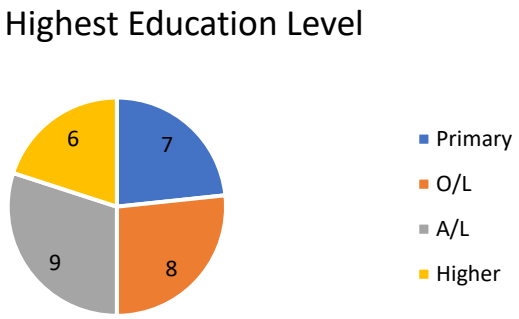
Quantitative data was analyzed using IBM-SPSS software for statistical computing. Descriptive statistics, frequencies, and Pearson correlation matrices were generated to identify relationships between variables, such as the correlation between technology acceptance and smartwatch support. Qualitative data underwent a rigorous thematic analysis, involving open coding and categorization, to extract core themes regarding patient vulnerability and system requirements.

Demographic Profile of Quantitative Data

The quantitative sample consisted of 30 respondents. The gender distribution was closely balanced, though slightly skewing male. Participants represented a mature demographic, with the highest frequency of respondents falling into the 41-50 age bracket, followed by the 51-60 group, and a smaller segment aged over 60. Educational backgrounds varied, with the largest portion having completed A/L education, followed by O/L, primary education, and higher education.







Findings

The analysis of the data collected from 30 survey respondents and 10 in-depth interviews reveals a clear intersection between human vulnerability, communication barriers, and the urgent need for technological intervention.

Questionnaire Index:

Q1.1 Aware of early signs of cardiac emergencies

Q1.2 Can recognize conditions requires immediate medical attention

Q1.3 Delayed seeking medical help during emergency

Q1.4 Fear, confusion or uncertainty cause delaying seeking care

Q2.1 Transport difficulties delayed arrival to the hospital

Q2.2 Ambulance services not available immediately when needed

Q2.3 Delay in receiving treatments after reaching hospital

Q2.4 Quick response during cardiac emergencies can save lives

Q2.5 Delays in emergency care increase the risk of complication or death

Q3.1 There should be faster way to alert family during emergencies

Q3.2 There should be system to quickly notify hospital about cardiac emergencies

Q3.3 Immediate communication between patients, family and hospitals is important

Q3.4 Technology can help reduce delays in emergency response

Q4.1 Comfortable using wearable devices for health monitoring

Q4.2 Device monitors heart conditions would be useful

Q4.3 Automatic alerts during emergencies would improve patient's safety

Q4.4 Trust technology to assist in managing health emergencies

Q5.1 Smartwatch detects heart problems and send alerts would be useful

Q5.2 Willing to use a smartwatch for emergency health monitoring

Q5.3 Smartwatch that alerts family and hospitals can reduce response time

Q5.4 Smartwatch based alerts system can improve survival chances

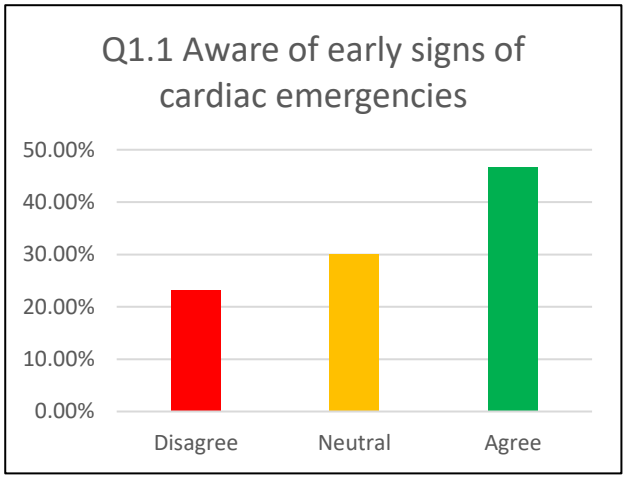
Q5.5 Recommend using such a smartwatch system for others

1. Symptom Awareness and Decision Delays

A significant factor contributing to emergency delays is the patient's own inability to recognize the severity of their symptoms. Quantitative analysis indicates mixed levels of awareness; while 46.7% of respondents agreed they are aware of the early signs of cardiac emergencies (Q1.1), a notable 53.3% were either neutral or disagreed. This uncertainty directly translates to delayed action.

Frequencies of Q1.1

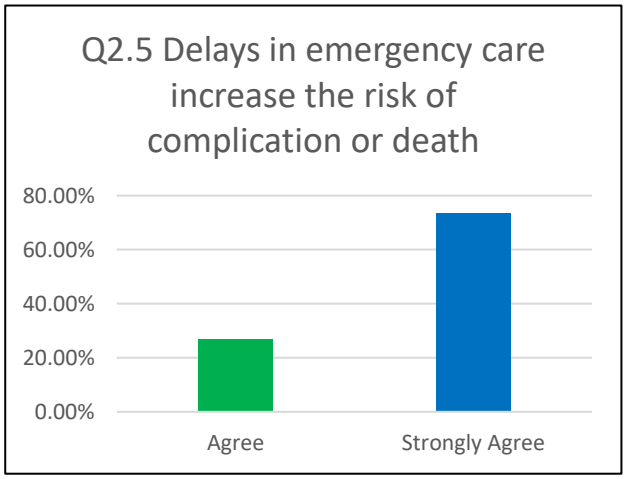
Q1.1	Counts	% of Total	Cumulative %
Disagree	7.00	23.3%	23.3%
Neutral	9.00	30.0%	53.3%
Agree	14.00	46.7%	100.0%



The thematic analysis strongly supports this. Theme 2 (Lack of Symptom Awareness Leads to Critical Delays) revealed that participants frequently mistook early warning signs for less severe conditions like gastric discomfort. This "wait and see" approach is dangerous, as 73.3% of survey respondents strongly agreed that delays in emergency care increase the risk of complication or death (Q2.5).

Frequencies of Q2.5

Q2.5	Counts	% of Total	Cumulative %
Agree	8.00	26.7%	26.7%
Strongly Agree	22.00	73.3%	100.0%

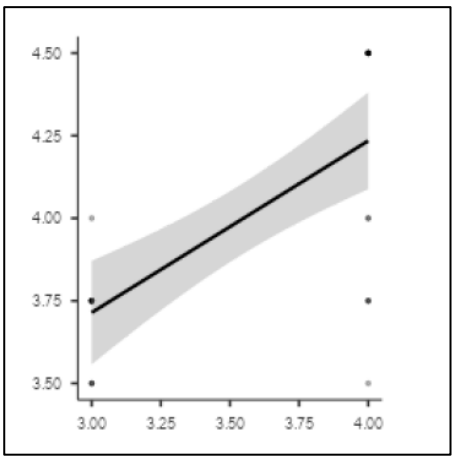


Furthermore, the Pearson Correlation matrix highlights a strong positive correlation ($r = 0.685$, $p < .001$) between fear/confusion (Q1.4) and delays in seeking treatment. The data definitively shows that when patients are confused about their symptoms, they hesitate, costing them critical minutes within the "golden hour."

Correlation Matrix

		Q1.4	Delay_score
Q1.4	Pearson's r	—	
	df	—	
	p-value	—	
Delay_score	Pearson's r	0.685	—
	df	28	—
	p-value	<.001	—

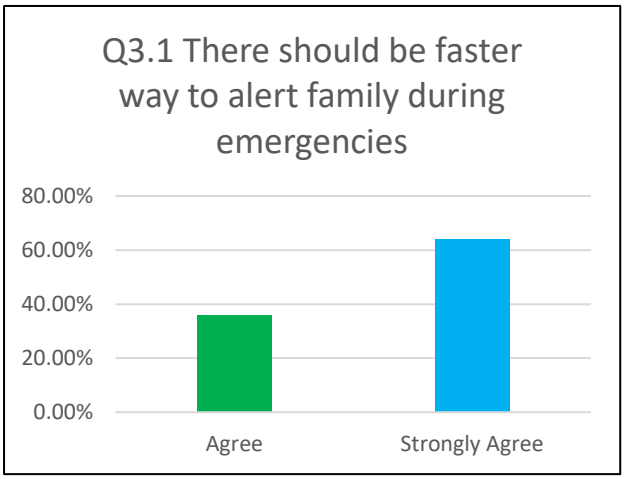
Note. H_a is positive correlation

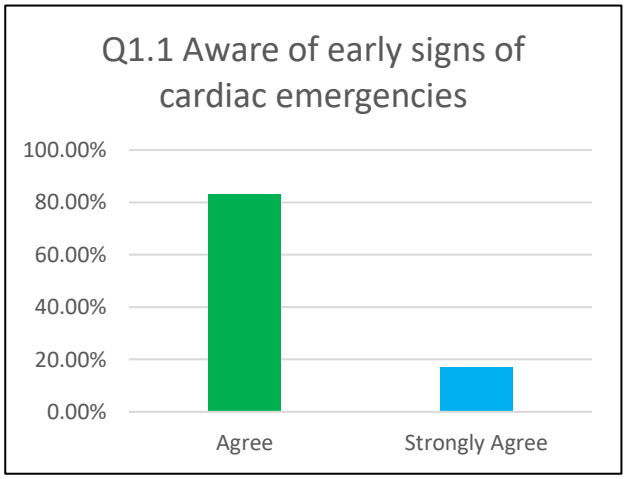


2. Communication Barriers and Vulnerability

Beyond symptom awareness, physical and logistical barriers severely hinder response times. Qualitative interviews generated Theme 3 (Communication Limitations During Emergencies) and Theme 4 (High Vulnerability of Individuals Living Alone). Participants reported being physically weakened during an attack, rendering them unable to manually dial a phone or explain their condition to a dispatcher.

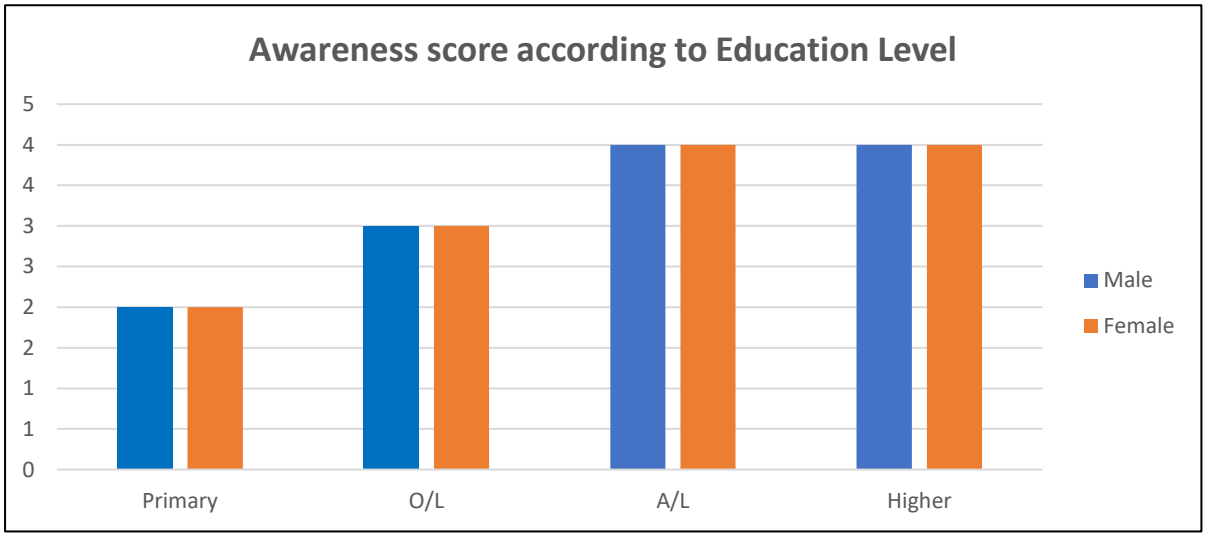
This dependence on others is a critical flaw in current emergency response protocols, particularly for the elderly or those living alone. Social isolation drastically increases mortality risk because the traditional "reactive" emergency chain is broken if the patient cannot initiate the call.





3. The Impact of Demographics on Awareness

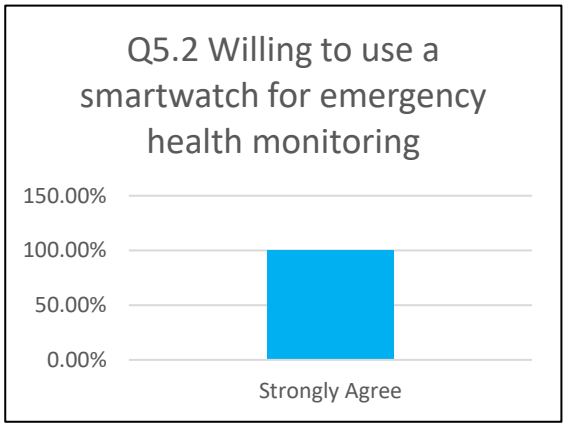
Demographic cross-tabulation reveals that education plays a role in symptom recognition. The bar plot analyzing Awareness according to Education Level demonstrates that patients with Higher Education or A/L qualifications consistently scored higher in recognizing emergency conditions compared to those with only Primary or O/L education. However, regardless of educational background, the physical inability to call for help remains a universal barrier during an actual cardiac event.



4. Acceptance of Wearable Technology and IoT Solutions

Given the identified delays, there is overwhelming support for a continuous, automated monitoring system. The quantitative data shows massive acceptance for smartwatch integration:

- 100% of respondents Strongly Agreed that they are willing to use a smartwatch for emergency health monitoring (Q5.2).



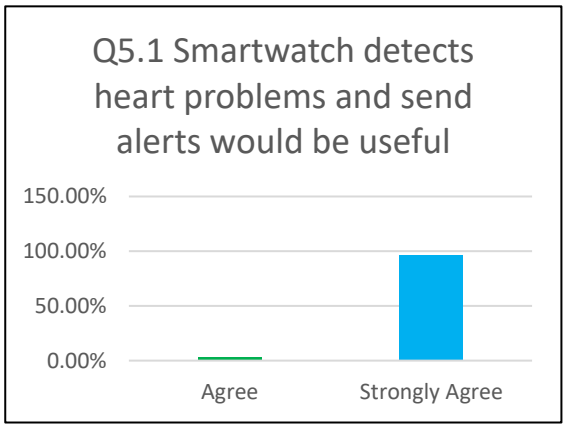
Frequencies of Q5.2

Q5.2	Counts	% of Total	Cumulative %
Strongly Agree	30.0	100.0%	100.0%

- 96.7% Strongly Agreed that a smartwatch detecting heart problems and sending alerts would be useful (Q5.1).

Frequencies of Q5.1

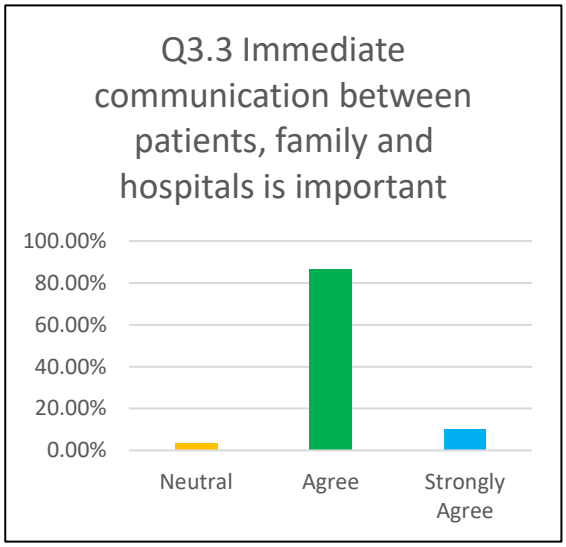
Q5.1	Counts	% of Total	Cumulative %
Agree	1.00	3.3%	3.3%
Strongly Agree	29.00	96.7%	100.0%



- Almost 100% Agreed or Strongly Agreed that immediate communication between patients, family, and hospitals is important (Q3.3).

Frequencies of Q3.3

Q3.3	Counts	% of Total	Cumulative %
Neutral	1.00	3.3%	3.3%
Agree	26.00	86.7%	90.0%
Strongly Agree	3.00	10.0%	100.0%

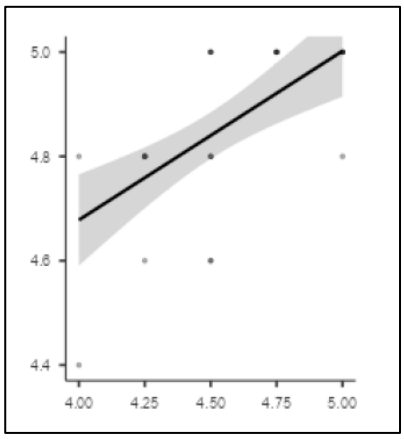


The Pearson Correlation Matrix solidifies this by showing a strong positive correlation ($r = 0.640$, $p < .001$) between general technology acceptance and specific smartwatch acceptance. Qualitative data from Theme 6 echoes this enthusiasm, showing that patients trust wearable devices to provide the continuous monitoring and automatic family notification they desperately need.

Correlation Matrix

		Tech_score	Watch_score
Tech_score	Pearson's r	—	
	df	—	
	p-value	—	
Watch_score	Pearson's r	0.640	—
	df	28	—
	p-value	<.001	—

Note. H_a is positive correlation



Ethical Considerations

This study was conducted with strict adherence to ethical research standards, prioritizing the dignity, privacy, and well-being of all participants. Prior to data collection, all respondents and interviewees were fully informed about the purpose of the research, the nature of their involvement, and their right to withdraw from the study at any time without consequence. Explicit informed consent was obtained from each participant.

Given the medical nature of the topic, conversations regarding cardiac emergencies and personal vulnerability were handled with high sensitivity to prevent any emotional distress. All quantitative and qualitative data were anonymized, securely stored, and strictly utilized solely for the academic purposes of this research.

Final Discussion

The findings of this study establish a comprehensive understanding of why cardiac patients in the Kandy district experience life-threatening delays in emergency care. The integration of quantitative survey data and qualitative thematic analysis reveals a dangerous overlap between human vulnerability, misinterpretation of symptoms, and systemic communication failures.

A primary obstacle identified is the delay in patient decision-making. The data clearly shows that when individuals are uncertain about their symptoms or mistake them for minor ailments, they hesitate to seek medical help. This hesitation is severely compounded when a cardiac event physically incapacitates a patient, leaving them unable to manually call for an ambulance or notify family members. This physical communication barrier makes elderly populations and those living alone exceptionally vulnerable, breaking the traditional "reactive" emergency response chain. Furthermore, the geographical and traffic-related logistical challenges unique to the Kandy area mean that any delay in dispatching an ambulance drastically diminishes the "golden hour" window for survival.

Despite these critical challenges, the study uncovers a highly promising avenue for intervention: the overwhelming acceptance of wearable health technology. Regardless of age or educational background, participants demonstrated deep trust in smart devices to manage health emergencies. The strong positive correlation between technology acceptance and the desire for automated emergency alerts indicates that patients are more than ready to transition from manual, reactive emergency calls to proactive, automated continuous monitoring. The findings conclusively point to an urgent need for an integrated technological ecosystem that bypasses human hesitation and physical limitations entirely.

Recommendations

Based on the research findings, the following strategies are proposed to drastically reduce emergency response times and improve cardiac patient survival rates in the Kandy district:

1. Implementation of a Smartwatch-Based IoT Monitoring System

The primary recommendation is the deployment of a continuous, 24/7 cardiac monitoring smartwatch tailored for high-risk patients. This wearable device must be equipped with advanced sensors to track heart rate variability, detect arrhythmias, and identify signs of cardiac

arrest. By shifting from intermittent check-ups to continuous physiological tracking, the system can detect critical anomalies before the patient even recognizes the symptoms.

2. Development of an Automated Dual-Alert Protocol

To eliminate the communication barriers identified in the study, the smartwatch must feature an automated emergency trigger. If vitals cross a dangerous threshold, or if the patient becomes unresponsive, the device should automatically and simultaneously send a GPS-tagged alert to both the patient's designated family members (via a mobile app) and the nearest emergency medical service. This cuts the "information chain" down to zero seconds.

3. Integration of a Centralized Hospital Web Interface

To improve emergency room preparedness, the smartwatch system must sync with a dedicated web interface accessible by hospital staff. This dashboard should feature a modern, clean, and minimal design aesthetic to ensure that medical professionals can instantly interpret incoming real-time ECG and heart rate data without visual clutter. Providing doctors with this continuous data stream allows them to prepare the necessary interventions before the ambulance even arrives at the hospital doors.

4. Community Awareness and Digital Literacy Campaigns

While technology provides the safety net, public health initiatives must still address the root cause of symptom ignorance. Healthcare providers should conduct awareness campaigns educating the public on recognizing silent or atypical cardiac symptoms. Additionally, onboarding programs should be established to help older demographics feel comfortable operating and charging their smartwatches, ensuring consistent daily use.

Limitations of the Study

While this study provides highly actionable insights into cardiac emergency response in the Kandy district, it is subject to certain limitations. First, the sample size consisting of 30 quantitative survey respondents and 10 qualitative interviewees is relatively small. While sufficient for identifying strong correlations and primary themes within a specific demographic, the findings may not be entirely generalizable to the entire population of Sri Lanka.

Second, the study focuses specifically on the geographical and logistical context of the Kandy district. Regions with vastly different infrastructure or emergency medical service (EMS) capabilities may experience different delay factors. Furthermore, the data relies heavily on self-reported patient experiences, which can be subject to recall bias, especially when participants are describing highly stressful or traumatic health emergencies. Finally, while the study establishes strong patient acceptance for a smartwatch and web interface system, it does not evaluate the technical or financial readiness of local hospitals to integrate such a centralized dashboard into their current IT infrastructure.

Conclusion

This research demonstrates that delays in emergency cardiac care in the Kandy district are driven by a complex intersection of human hesitation, physical vulnerability, and logistical barriers. The traditional, reactive model of emergency response where a patient must actively recognize their symptoms and manually call for help is critically flawed when dealing with severe cardiovascular events. As evidenced by the data, lack of symptom awareness and the physical incapacitation that occurs during a heart attack leave patient, particularly the elderly and those living alone, dangerously exposed.

However, the findings also reveal a clear and highly accepted path forward. There is overwhelming public willingness to adopt wearable IoT technology to bridge these gaps. Patients strongly trust and desire a system that can bypass their own uncertainties and physical limitations.

The proposed solution a continuous 24/7 monitoring smartwatch linked to an automated family and hospital web interface directly addresses the core problems identified in this study. By replacing manual emergency calls with automated, real-time physiological alerts, this technology can eliminate decision delays, overcome communication barriers, and allow hospitals to prepare for incoming patients before they arrive. Ultimately, integrating this smart ecosystem is not just a technological enhancement; it is a vital, life-saving necessity capable of maximizing the "golden hour" and significantly reducing cardiac mortality rates in the Kandy area.

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6. Quantitative data collected through Questionnaire and Qualitative data collected through Interviews.

Appendices

1. Questionnaire (Sinhala)

හෘද රෝගීන් සඳහා හදිසි අවස්ථාවකදී සෞඛ්‍ය සේවා ලබාගැනීමට බලපාන සාධක සහ වේගවත් ප්‍රතිකාර ක්‍රම පිළිබඳ අධ්‍යනයක්

ගරු මහත්මයා/මහත්මියනි,

මම නෙත්මි නිමේෂා (ජා.හැ. 200278302631) පේරාදෙණිය විශ්වවිද්‍යාලයේ ශාස්ත්‍ර පීඨයේ, සමාජ විද්‍යා අධ්‍යයන අංශයේ සමාජවෘතවේදී (ගෞරව) උපාධියේ තෙවන වසර සිසුවියක් වෙමි. හෘද රෝගීන් සඳහා හදිසි අවස්ථාවකදී සෞඛ්‍ය සේවා ලබාගැනීමට බලපාන සාධක සහ වේගවත් ප්‍රතිකාර ක්‍රම පිළිබඳ සොයා බැලීමේ අරමුණින් පේරාදෙණිය ලියෝ සමාජය නියෝජනය කරමින් මෙම පර්යේෂණය සිදු කරනු ලැබේ. ඔබගේ සහභාගීත්වය සම්පූර්ණයෙන්ම ස්වේච්ඡාවෙන් සිදු වූවද, මෙම පර්යේෂණය සඳහා ඔබගේ දායකත්වය ඉතා වැදගත් වේ. ඔබ ලබා දෙන තොරතුරු නිර්නාමික වන අතර, කිසිදු අවස්ථාවක කිසිදු තෙවන පාර්ශවයකින් හඳුනා නොගැනීමට ඔබට සහතික වෙමි. මෙම අධ්‍යයනය සම්බන්ධයෙන් ඔබට යම් ගැටලුවක් වේ නම් පහත දුරකථන අංකය ඔස්සේ මා සම්බන්ධ කර ගැනීමට පසුබට නොවන්න. ස්තූතියි!

දු.ක. : 077 043 8560

A කොටස: පෞද්ගලික තොරතුරු

1. වයස (අවුරුදු)

☐ **40** ට අඩු ☐ **41 - 50** අතර ☐ **51 - 60** අතර ☐ **60** ට වැඩි

2. ස්ත්‍රී පුරුෂ භාවය

☐ ස්ත්‍රී ☐ පුරුෂ ☐ වෙනත්

3. ඉහළම අධ්‍යාපන මට්ටම

☐ ප්‍රාථමික ☐ සාමාන්‍ය පෙළ ☐ උසස් පෙළ ☐ උසස් අධ්‍යාපනය

4. රැකියාව : _____

5. දිස්ත්‍රික්කය : _____

6. ඔබ මීට පෙර හෘදය ආශ්‍රිත දුර්වලතාවකට ලක්වී තිබේද? ☐ ඔව් ☐ නැත

B කොටස: ප්‍රකාශ (එක් විකල්පයක් ලකුණු කරන්න)

කරුණාකර එක් එක් ප්‍රකාශයට වඩාත් සුදුසු අංකය සළකුණු කිරීමෙන් ඔබේ ප්‍රතිචාරය දක්වන්න. මෙම පරිමාණය භාවිතා කරන්න. (1 - සම්පූර්ණයෙන් එකග නොවෙමි, 2 - එකග නොවෙමි, 3 - මැදිහත්, 4 - එකග වෙමි, 5 - සම්පූර්ණයෙන් එකග වෙමි)

	1 කොටස: දැනුවත්භාවය සහ මූලික ප්‍රතිචාරය	1	2	3	4	5
1.1	හෘද රෝගී හදිසි අවස්ථාවක පූර්ව අනතුරු ඇඟවීමේ සලකුණු පිළිබඳව මම දැනුවත් වෙමි.					
1.2	මගේ තත්ත්වය සඳහා වහාම වෛද්‍ය ප්‍රතිකාර අවශ්‍ය වන්නේ කවදාදැයි මට හඳුනාගත හැකිය.					
1.3	මගේ හදිසි අවස්ථාවේදී වෛද්‍ය ප්‍රතිකාර ලබා ගැනීමට මම ප්‍රමාද වීම් සිදු විය.					
1.4	බිය, ව්‍යාකූලත්වය හෝ අවිශ්වාසය ප්‍රතිකාර සෙවීමේ ප්‍රමාදයන්ට හේතු වේ.					
	2 කොටස: හදිසි ප්‍රවේශය සහ ප්‍රමාදයන්					
2.1	ප්‍රවාහන දුෂ්කරතා රෝහලට පැමිණීම ප්‍රමාද වීමට හේතු වේ.					
2.2	අවශ්‍ය අවස්ථාවලදී ගිලන් රථ සේවා වහාම ලබාගත නොහැකි වේ.					
2.3	රෝහලට පැමිණීමෙන් පසුවත් ප්‍රතිකාර ලබා ගැනීමට ප්‍රමාදයක් සිදු වේ.					
2.4	හෘද රෝගී හදිසි අවස්ථාවකදී ඉක්මන් ප්‍රතිචාර දැක්වීමෙන් ජීවිත බේරා ගත හැකිය.					
2.5	හදිසි ප්‍රතිකාර ප්‍රමාද වීමෙන් සංකූලතා ඇතිවීමේ හෝ මරණයට පත්වීමේ අවදානම වැඩි වේ.					
	3 කොටස: වේගවත් ප්‍රතිචාර පද්ධතියක අවශ්‍යතාවය					
3.1	හදිසි අවස්ථාවකදී පවුලේ සාමාජිකයන්ට දැනුම් දීම සඳහා වඩාත් වේගවත් ක්‍රමයක් තිබිය යුතුය.					
3.2	හෘද රෝගී හදිසි අවස්ථා පිළිබඳව රෝහල්වලට ඉක්මනින් දැනුම් දීමට පද්ධතියක් තිබිය යුතුය.					
3.3	රෝගීන්, පවුලේ අය සහ රෝහල් අතර කඩිනම් සන්නිවේදනය වැදගත් වේ.					
3.4	හදිසි ප්‍රතිචාර දැක්වීමේ ප්‍රමාදයන් අවම කිරීමට තාක්ෂණයට උදවු කළ හැකිය.					
	4 කොටස: තාක්ෂණය භාවිතය					
4.1	සෞඛ්‍යය නිරීක්ෂණය කිරීම සඳහා පැළඳිය හැකි (Wearable) උපාංග භාවිතා කිරීමට මම කැමැත්තෙමි.					
4.2	හදවතේ ක්‍රියාකාරීත්වය අඛණ්ඩව නිරීක්ෂණය කරන උපාංගයක් ප්‍රයෝජනවත් වනු ඇත.					
4.3	හදිසි අවස්ථාවකදී ස්වයංක්‍රීයව දැනුම් දීම් සිදු කරන ක්‍රමවේදයක් මඟින් රෝගී ආරක්ෂාව වැඩි දියුණු කළ හැකිය.					
4.4	හදිසි සෞඛ්‍ය තත්ත්වයන් පාලනය කිරීමට උදවු වන තාක්ෂණය පිළිබඳව මට විශ්වාසයක් ඇත.					
	5 කොටස: ස්මාර්ට් වොච් (Smart Watch) පදනම් කරගත් විසඳුම					
5.1	හෘද ගැටලු හඳුනාගෙන අනතුරු ඇඟවීම් යවන ස්මාර්ට් වොච් එකක් ප්‍රයෝජනවත් වනු ඇත.					
5.2	හදිසි සෞඛ්‍ය නිරීක්ෂණ කටයුතු සඳහා ස්මාර්ට් වොච් එකක් භාවිතා කිරීමට මම කැමැත්තෙමි.					
5.3	පවුලේ සාමාජිකයන්ට සහ රෝහල්වලට එකවර දැනුම් දෙන ස්මාර්ට් වොච් පද්ධතියකට ප්‍රතිචාර දැක්වීමේ කාලය අඩු කළ හැකිය.					
5.4	ස්මාර්ට් වොච් පදනම් කරගත් අනතුරු ඇඟවීමේ පද්ධතිවලට හෘද රෝගී හදිසි අවස්ථාවලදී ජීවිත බේරා ගැනීමේ අවස්ථාව වැඩි කළ හැකිය.					
5.5	මෙවැනි ස්මාර්ට් වොච් පද්ධතියක් භාවිතා කරන ලෙස මම අනෙක් අයටද නිර්දේශ කරමි.					

ස්තූතියි!

2. Interview Guideline (Sinhala)

Interview Guideline in Sinhala

මම නෙත්මි නිමේෂා (ජා.හැ. **200278302631)** පේරාදෙණිය විශ්වවිද්‍යාලයේ ශාස්ත්‍ර පීඨයේ, සමාජ විද්‍යා අධ්‍යයන අංශයේ සමාජවෘතවේදී (ගෞරව) උපාධියේ තෙවන වසර සිසුවියක් වෙමි. හෘද රෝගීන් සඳහා හදිසි අවස්ථාවකදී සෞඛ්‍ය සේවා ලබාගැනීමට බලපාන සාධක සහ වේගවත් ප්‍රතිකාර ක්‍රම පිළිබඳ සොයා බැලීමේ අරමුණින් පේරාදෙණිය ලියෝ සමාජය නියෝජනය කරමින් මෙම පර්යේෂණය සිදු කරනු ලැබේ. ඔබගේ සහභාගීත්වය සම්පූර්ණයෙන්ම ස්වේච්ඡාවෙන් සිදු වූවද, මෙම පර්යේෂණය සඳහා ඔබගේ දායකත්වය ඉතා වැදගත් වේ. ඔබ ලබා දෙන තොරතුරු නිර්නාමික වන අතර, කිසිදු අවස්ථාවක කිසිදු තෙවන පාර්ශවයකින් හඳුනා නොගැනීමට ඔබට සහතික වෙමි.

සම්මුඛ සාකච්ඡාව ආරම්භ කිරීමට පෙර ඔබට කිසියම් ප්‍රශ්ණ තිබේද? නැතිනම් සම්මුඛ සාකච්ඡාව ඉදිරියට කරගෙන යාමට ඔබ කැමතිද? ඔබගේ අවසරය ඇතිව මෙම සංවාදය පටිගත කෙරේ.

A: පසුබිම් තොරතුරු

- කරුණාකර ඔබ ගැන කෙටියෙන් කියන්න පුළුවන්ද? (වයස, රැකියාව, සාමාන්‍ය සෞඛ්‍ය තත්ත්වය)
- මීට පෙර ඔබට හදිසි හෘදයාබාධ තත්ත්වයකට මුහුණ දීමට සිදු වී තිබේද? එසේ නම්, එය සිදු වූයේ කවදාද?

B: හදිසි අවස්ථාවේදී ලැබූ අත්දැකීම

- රෝග ලක්ෂණ මුලින්ම දැනුණු විට කුමක් සිදුවූවාද යන්න විස්තර කළ හැකිද?
- එම අවස්ථාවේදී ඔබට ඇතිවූ රෝග ලක්ෂණ මොනවාද?
- රෝග ලක්ෂණ ආරම්භ වූ විට ඔබ ඊට ප්‍රතිචාර දැක්වූයේ කෙසේද?

C: වෛද්‍ය ආධාර ලබාගැනීමට ගත් තීරණය

- ඔබ වහාම වෛද්‍ය ආධාර ලබා ගත්තාද? එසේ කළේ හෝ නොකළේ ඇයි?
- රෝහලට යාම ප්‍රමාද වීමට බලපෑ යම් හේතු තිබුණාද?
- වෛද්‍ය ප්‍රතිකාර ලබා ගැනීමට තීරණය කිරීමට ඔබට උපකාර කළේ කවුද? (පවුලේ අය, මිතුරන්, ඔබගේම තීරණයක්ද?)

D: ප්‍රවාහනය සහ ප්‍රතිකාර සඳහා ප්‍රවේශය

- ඔබ රෝහලට ගියේ කෙසේද? (ගිලන් රථයක්, පෞද්ගලික වාහනයක්, යනාදිය)
- ප්‍රවාහන පහසුකම් ලබා ගැනීමේදී ඔබ යම් අපහසුතාවලට මුහුණ දුන්නාද?
- රෝහලට ළඟා වීමට කොපමණ වේලාවක් ගතවුණාද?
- ඔබේ අදහසට අනුව, ප්‍රවාහන ප්‍රමාදයන් ඔබේ තත්ත්වයට බලපෑමක් කළාද?

E: රෝහලේ අත්දැකීම

- ඔබ රෝහලට පැමිණි විට කුමක් සිදුවුණාද?
- ඔබට වහාම ප්‍රතිකාර ලැබුණාද, නැතහොත් යම් ප්‍රමාදයක් තිබුණාද?
- ඔබට ලබා දුන් හදිසි ප්‍රතිකාර ගැන ඔබ කොපමණ දුරට සෑහීමකට පත් වෙනවාද?

F: සන්නිවේදනය සහ දැනුවත්භාවය

- ඔබේ පවුලේ සාමාජිකයන් ඔබේ තත්ත්වය ගැන වහාම දැනුවත් වුණාද?
- වඩාත් හොඳ සන්නිවේදනයක් තිබුණා නම් ඔබට ඉක්මනින් ප්‍රතිකාර ලබා ගැනීමට හැකි වනු ඇතැයි ඔබ සිතනවාද?
- මෙම සිදුවීමට පෙර, හදිසි හෘද රෝග ලක්ෂණ ගැන ඔබ කොතරම් දුරට දැනුවත්ව සිටියාද?

G: ප්‍රමාදයන් පිළිබඳව ආවර්ජනය කිරීම

- ආපසු හැරී බලන විට, ප්‍රතිකාර ලබා ගැනීමේදී සිදු වූ විශාලතම ප්‍රමාදයට හේතුව කුමක් යැයි ඔබ සිතනවාද?
- ඔබට වඩාත් ඉක්මනින් වෛද්‍ය ප්‍රතිකාර ලබා ගැනීමට උපකාරී වන්නට තිබුණේ කුමක්ද?

H: තාක්ෂණය සහ ස්මාර්ට් ඔරලෝසු **(Smartwatch)** අදහස

- ඔබ කවදා හෝ සෞඛ්‍ය නිරීක්ෂණ උපාංගයක් (ස්මාර්ට් ඔරලෝසුවක් හෝ ෆිට්නස් බෑන්ඩ් එකක් වැනි) භාවිතා කර තිබේද?
- හෘදයේ ක්‍රියාකාරිත්වය අඛණ්ඩව නිරීක්ෂණය කරන උපාංග ගැන ඔබ සිතන්නේ කුමක්ද?
- හෘද ගැටළු හඳුනාගෙන ස්වයංක්‍රීයව අනතුරු ඇඟවීම් යවන ස්මාර්ට් ඔරලෝසුවක් හදිසි අවස්ථාවලදී උපකාරී වනු ඇතැයි ඔබ සිතනවාද?
- එවැනි උපාංගයකට ඔබේ පවුලේ සාමාජිකයින්ට වහාම අනතුරු ඇඟවීමට හැකි නම් එය කොතරම් ප්‍රයෝජනවත් වේවිද?
- එවැනි උපාංගයක් භාවිතා කිරීමට ඔබ පහසුවක්/කැමැත්තක් දක්වනවාද? එසේ වීමට හෝ නොවීමට හේතුව කුමක්ද?

I: යෝජනා

- හදිසි හෘදයාබාධ ප්‍රතිකාරවල ප්‍රමාදයන් අවම කිරීම සඳහා ඔබ යෝජනා කරන වැඩිදියුණු කිරීම් මොනවාද?
- වඩා වේගවත් ප්‍රතිචාරයක් සඳහා තාක්ෂණය මත පදනම් වූ විසඳුම් හඳුන්වා දීම ගැන ඔබ සිතන්නේ කුමක්ද?

ස්තූතියි!